

بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ

NASAB

The Global Shajra Nasab Platform

Full Product & Technical Architecture

Confidential — Internal Planning Document

Version 1.0 · 2026

1. Vision & Mission

Nasab is a non-profit, public-interest digital platform dedicated to preserving, verifying, and visualising the blessed lineage of the Prophet Muhammad ﷺ — and connecting the millions of living descendants worldwide through a single, trustworthy, open source of truth.

1.1 Mission Statement

To build the world's most authoritative, inclusive, and beautifully designed Shajra Nasab — free forever, open to all scholarly traditions, and governed by scholarly integrity.

1.2 Core Principles

- Non-discriminatory — all branches, all scholarly traditions (Sunni and Shia) presented with equal rigour
- Source-cited — every record annotated with its scholarly or documentary source
- Verification-first — no lineage claim is published without multi-source corroboration
- Non-profit forever — no paywalls, no ads, institutionally partnered
- Bilingual at core — Arabic and English, with RTL/LTR full support

1.3 Target Audience

- Sayyids and Sharifs worldwide seeking to document and verify their lineage
- Islamic scholars, genealogists, and historians
- General Muslim public with educational interest
- Islamic institutions, Maqams, and royal houses (Hashemites etc.)
- Academic researchers in Islamic history and genealogy

2. Product Specification

2.1 Core Features at Launch

2.1.1 Interactive Family Tree Visualisation

- Collapsible node graph — expand/collapse individual branches
- Zoom and pan — smooth touch and mouse navigation
- Filter by branch — Hasanid (Sharif), Husaynid (Sayyid), Hashemite, all sub-lines
- Search by name — Arabic or English, fuzzy matching, instant results
- Node biography panel — click any person to view full detail card
- Mobile fully responsive — works on all screen sizes
- Download & share tree image — PNG export of visible tree state

2.1.2 Biographical Detail Panel

- Full name in Arabic script (canonical) + English transliteration
- Birth and death dates with hijri/gregorian dual display
- Title and honorifics (Imam, Sharif, Sayyid, etc.)
- Generation number from the Prophet ﷺ
- Branch classification (Hasanid / Husaynid / sub-line)
- Scholarly tradition notes (Sunni and/or Shia sources cited)
- Key historical significance summary
- Source citations for all data

2.1.3 User Account & Lineage Submission

- Free account creation (email + password, or social login)
- Users submit their personal shajra connecting to the verified historical tree
- Upload supporting documents: physical shajra (PDF/image), official certificates
- Submission enters a verification queue — not published until approved
- Users can view their submission status in a personal dashboard
- Private lineages — user controls who can see their branch

2.1.4 Verification Workflow

- Each submission reviewed by admin and/or appointed scholarly verifier
- Any one of the following accepted as proof: physical shajra document, Official Sayyid/Sharif certificate from recognised authority, community elder or scholar attestation, or multiple corroborating sources

- Verifier annotates the record with proof type and source
- Rejected submissions receive written reason and can be resubmitted with additional evidence
- Approved records are published with a verified badge and source citation

2.1.5 Admin Dashboard

- Full tree management — add, edit, delete any node
- Submission queue — review, approve, reject, request more info
- User management — roles, bans, verifier appointments
- Data export — full tree as JSON or CSV
- Audit log — every change tracked with timestamp and actor
- Analytics — visitor stats, submission rates, popular branches

2.1.6 Urdu PDF Import (Phase 1 Priority)

- Your handwritten Urdu shajra PDF will be digitised and structured
- Process: PDF upload → OCR (Google Vision or AWS Textract for Nastaliq Urdu) → manual correction → structured JSON → imported into database
- Claude will guide you through this process step by step

2.2 Language & Localisation

- Arabic — primary script for all names; UI fully RTL
- English — full UI translation; LTR layout
- Language toggle in header — persists via user preference
- All names stored canonically in Arabic; English is a transliteration layer
- Additional languages (Urdu, Persian, Turkish) in Phase 2

2.3 Scholarly Integrity Model

Where Sunni and Shia sources agree, a single record is displayed. Where they differ, both traditions are shown side-by-side within the biography panel, each with its source cited. No tradition is privileged over another. The tree presents historical reality as documented across the full breadth of Islamic scholarship.

3. Brand & Identity

3.1 Recommended Brand Name

NASAB • نَسَب

'Nasab' (نَسَب) is the classical Arabic word for lineage and descent — precise, universally understood across all Muslim communities worldwide, modern enough for a global digital platform, and deeply rooted in the Islamic scholarly tradition. It requires no explanation to the target audience in any language.

3.2 Suggested Domain

- nasab.org — ideal (non-profit .org, globally recognised)
- nasab.app — modern alternative
- shajranasab.org — more descriptive if nasab.org is unavailable

3.3 Design Direction

- Aesthetic: Refined Islamic manuscript — deep midnight navy, aged gold, geometric arabesque accents
- Typography: Classical Arabic calligraphic display font + clean modern body font
- Tone: Scholarly, trustworthy, reverent — never commercial
- Logo concept: Stylised tree or chain motif using geometric Islamic patterns
- Colour palette: Navy (#0D1B2A), Gold (#C9A84C), Cream (#F5ECD7), Deep Green (#1B5E20)

4. Recommended Technology Stack

4.1 Stack Decision & Rationale

Given that you have no deadline, want to do this properly, and need Claude to guide everything — the following stack is recommended for its maturity, developer community, Vercel-native deployment, and suitability for a bilingual RTL/LTR tree application.

Layer	Technology	Rationale
Frontend Framework	Next.js 14 (App Router)	React-based, full-stack, Vercel-native, excellent i18n support
Tree Visualisation	React Flow + D3.js	Production-grade graph rendering, zoom/pan, collapsible nodes
Styling	Tailwind CSS + shadcn/ui	Utility-first, RTL plugin available, accessible components
Language / i18n	next-intl	Best-in-class Next.js i18n, RTL/LTR switching
Backend / API	Next.js API Routes + tRPC	Type-safe API, no separate backend server needed
Database	Supabase (PostgreSQL)	Managed Postgres, built-in auth, file storage, realtime, free tier
Authentication	Supabase Auth	Email, social login, admin roles — all built in
File Storage	Supabase Storage	For uploaded shajra PDFs and certificates
Email Notifications	Resend	Modern transactional email, generous free tier
OCR (Urdu PDF)	Google Cloud Vision API	Best Nastaliq Urdu OCR available
Deployment	Vercel	Zero-config Next.js deployment, free tier, global CDN
Analytics	Vercel Analytics + Plausible	Privacy-respecting, no cookie banners needed
Image Export	html-to-image / Konva	Client-side PNG export of tree state

4.2 Why Supabase over other databases

- PostgreSQL — relational structure is perfect for hierarchical genealogical data
- Built-in auth — no need for separate Auth0 or Clerk subscription
- File storage — upload and serve shajra PDFs and certificates natively
- Row-level security — users can only see/edit their own records

- Realtime subscriptions — admin dashboard updates live as submissions arrive
- Generous free tier — suitable for launch phase with institutional partners

5. Database Schema

5.1 Core Tables

persons — the tree nodes

Column	Type	Description
id	UUID PK	Unique identifier
name_ar	TEXT NOT NULL	Canonical Arabic name (primary source of truth)
name_en	TEXT	English transliteration
father_id	UUID FK → persons	Parent node — forms the tree structure
mother_name_ar	TEXT	Mother's name (Arabic)
branch	ENUM	hasanid husaynid hashemite other
scholarly_tradition	ENUM	sunni shia both
birth_year_hijri	INTEGER	Birth year AH
death_year_hijri	INTEGER	Death year AH
birth_year_ce	INTEGER	Birth year CE
death_year_ce	INTEGER	Death year CE
title_ar	TEXT	Honorific title in Arabic
title_en	TEXT	Honorific title in English
generation	INTEGER	Generation from Prophet ﷺ (1 = Fatimah/Ali)
bio_ar	TEXT	Biography in Arabic
bio_en	TEXT	Biography in English
is_imam	BOOLEAN	True for the 12 Imams
is_verified	BOOLEAN DEFAULT true	Historical records auto-verified
sources	JSONB	Array of source citations
created_at	TIMESTAMPTZ	Record creation timestamp
updated_at	TIMESTAMPTZ	Last update timestamp

submissions — user lineage claims

Column	Type	Description
id	UUID PK	Unique identifier

Column	Type	Description
user_id	UUID FK → auth.users	Submitting user
claimed_ancestor_id	UUID FK → persons	Node they claim descent from
chain_ar	TEXT[]	Array of Arabic names in chain from ancestor to self
chain_en	TEXT[]	English transliteration of chain
generation_claimed	INTEGER	Their claimed generation number
proof_type	TEXT[]	Types of proof submitted
document_urls	TEXT[]	Supabase Storage URLs for uploaded documents
notes	TEXT	Additional notes from submitter
status	ENUM	pending under_review approved rejected more_info
reviewer_id	UUID FK → auth.users	Admin/verifier assigned
reviewer_notes	TEXT	Internal reviewer notes
rejection_reason	TEXT	Sent to user if rejected
submitted_at	TIMESTAMPTZ	Submission timestamp
reviewed_at	TIMESTAMPTZ	Review completion timestamp

users — extended profile

Column	Type	Description
id	UUID PK FK → auth.users	Matches Supabase auth user
full_name_ar	TEXT	Full name in Arabic
full_name_en	TEXT	Full name in English
country	TEXT	Country of residence
role	ENUM	user verifier admin superadmin
is_verified_descendant	BOOLEAN	True after lineage approved
approved_person_id	UUID FK → persons	Their node in the tree once approved
profile_visibility	ENUM	public verified_only private
created_at	TIMESTAMPTZ	Account creation

audit_log — change history

Column	Type	Description
id	UUID PK	Log entry ID
actor_id	UUID FK → auth.users	Who made the change
action	TEXT	create update delete approve reject
table_name	TEXT	Which table was affected
record_id	UUID	Which record was affected
diff	JSONB	Before/after values
created_at	TIMESTAMPTZ	Timestamp

6. Project Folder Structure

The following is the recommended Next.js 14 App Router project structure:

```

nasab/ └─ app/                                # Next.js App Router └─ [locale]/
# i18n routing (ar, en) └─ page.tsx             # Homepage with tree └─ ──
person/[id]/page.tsx # Individual biography page └─ search/page.tsx # Search
results └─ ── submit/page.tsx # Lineage submission form └─ ──
dashboard/page.tsx # User personal dashboard └─ admin/ # Admin area
└─ ── page.tsx # Admin overview └─ ── submissions/ # Review
queue └─ ── tree/ # Tree management └─ ── users/ #
User management └─ ── api/ # API routes └─ ── persons/ #
# Tree CRUD └─ ── submissions/ # Submission workflow └─ ── search/
# Search endpoint └─ components/ └─ tree/ # React Flow tree
components └─ ── FamilyTree.tsx # Main tree container └─ ── PersonNode.tsx
# Individual node card └─ ── BioPanel.tsx # Biography side panel └─ ──
TreeControls.tsx # Zoom, filter, search bar └─ ── ui/ #
shadcn/ui components └─ ── forms/ # Submission & auth forms └─ ──
layout/ # Header, Footer, Nav └─ lib/ └─ ── supabase/ #
DB client & queries └─ ── i18n/ # Translation files └─ ── ar.json
# Arabic strings └─ ── en.json # English strings └─ ── utils/
# Helpers, formatters └─ data/ └─ ── seed/ # Historical tree seed data
└─ ── persons.json # Structured JSON seed └─ public/ #
Static assets └─ ── middleware.ts # i18n routing middleware └─ supabase/
└─ migrations/ # SQL migration files └─ ── seed.sql #
Initial data seed

```

7. API Design

7.1 Public REST API (Third-Party Access)

All endpoints are read-only for public consumers. Authentication required for write operations. Rate-limited to 100 requests/minute per IP.

Method	Endpoint	Description	Auth
GET	/api/v1/persons	List persons (paginated, filterable)	Public
GET	/api/v1/persons/:id	Get single person with full biography	Public
GET	/api/v1/persons/:id/children	Get direct children of a node	Public
GET	/api/v1/persons/:id/ancestors	Get full ancestor chain to root	Public
GET	/api/v1/search?q=	Search by Arabic or English name	Public
GET	/api/v1/branches	List all branches with node counts	Public
GET	/api/v1/tree/root	Get tree from root (Prophet ﷺ) downward	Public
POST	/api/v1/submissions	Submit a lineage claim	User Auth
GET	/api/v1/submissions/:id	Get own submission status	User Auth
GET	/api/v1/admin/submissions	List all submissions (queue)	Admin
PATCH	/api/v1/admin/submissions/:id	Approve / reject / request info	Admin
POST	/api/v1/admin/persons	Add a new person to tree	Admin
PATCH	/api/v1/admin/persons/:id	Update a person record	Admin

7.2 API Response Format

```
{
  "success": true,
  "data": {
    "id": "uuid",
    "name_ar": "الحسن بن علي",
    "name_en": "Al-Hasan ibn Ali",
    "generation": 2,
    "branch": "hasanid",
    "scholarly_tradition": "both",
    "sources": ["Ibn Hazm, Jamharat Ansab al-Arab", "Al-Baladhuri, Ansab al-Ashraf"],
    "meta": { "version": "1.0", "timestamp": "2026-01-01T00:00:00Z" }
  }
}
```

8. Lineage Verification Workflow

8.1 Submission Flow

1. User creates account and logs in
2. User navigates to their claimed ancestor in the tree
3. User clicks 'Submit Your Lineage' on that node
4. User fills submission form: name chain in Arabic + English, generation number, any notes
5. User uploads at least one proof document (PDF, image)
6. Submission saved with status: pending — user receives confirmation email
7. Admin/verifier receives email notification of new submission
8. Verifier reviews documents, checks chain plausibility against existing tree
9. Verifier either: Approves (record published, user notified), Rejects with reason (user notified, can resubmit), or Requests more information (user notified)
10. Approved submissions create a new verified node in the tree linked to the claimed ancestor

8.2 Accepted Proof Types

- Physical shajra document (handwritten or printed, scanned as PDF/image)
- Official Sayyid or Sharif certificate issued by a recognised religious authority or Maqam
- Attestation letter from a community elder or Islamic scholar with their credentials
- Multiple corroborating sources (where no single document is available)

Any one of the above is sufficient. Verifiers may request a second source if the primary document raises questions.

8.3 Verifier Roles

- Verifiers are appointed by superadmin — initially you as the platform founder
- Verifiers can be Islamic scholars, genealogists, or trusted community figures
- Each approved record is tagged with verifier ID for accountability
- Verifiers cannot approve their own lineage submissions

9. Urdu PDF Digitisation Plan

Your handwritten Urdu shajra is the seed of your personal lineage branch. Here is the step-by-step plan to digitise it and integrate it into the platform.

9.1 Process

11. Share your PDF with Claude — we will analyse it together
12. Claude extracts the text using OCR (Google Cloud Vision — best available for Nastaliq Urdu)
13. Claude produces a structured JSON representation of every person, their father, and chain
14. You review and correct the JSON — Claude guides you through any ambiguous names
15. The corrected JSON is imported into the database via an admin import script
16. Your lineage chain is connected to the correct historical ancestor node in the main tree
17. Your record becomes the first user-submitted verified lineage on the platform

9.2 Expected Challenges

- Nastaliq Urdu handwriting varies significantly — OCR accuracy may be 70–85%, requiring manual correction
- Names may use regional or colloquial Arabic transliterations — Claude will help standardise to canonical Arabic
- Chain breaks or missing generations — Claude will flag these for you to resolve
- Patronymic vs. given name disambiguation — will be handled case by case

9.3 Output Format

```
[ { "name_ar": "محمد بن احمد", "name_en": "Muhammad ibn Ahmad",  
  "father_name_ar": "احمد بن حسن", "generation_from_prophet": 38, "notes":  
  "Born in Lahore, migrated to Karachi" }, ... ]
```

10. Build Phases

There is no deadline — each phase is completed properly before moving to the next.

Phase	Name	Key Deliverables
Phase 0	Foundation	Supabase project setup, schema migrations, seed historical data (Prophet 12 → ﷺ Imams → major branches → Hashemites), Next.js project scaffold, i18n (Arabic + English), Vercel deployment pipeline
Phase 1	Tree MVP	Interactive collapsible tree, zoom/pan, node biography panel, search, branch filter, mobile responsive, PNG export
Phase 2	Urdu PDF Import	Digitise your handwritten shajra, build import tooling, connect your lineage to the historical tree
Phase 3	User Accounts	Supabase auth, user registration, personal dashboard, submission form, document upload
Phase 4	Verification System	Admin dashboard, submission review queue, verifier roles, email notifications via Resend, audit log
Phase 5	Public API	RESTful public API with documentation, rate limiting, API key management for institutional partners
Phase 6	Institutional Partnerships	Partner onboarding flow, co-branded pages for partnering Islamic institutions, bulk data import tools
Phase 7	Scale & Polish	Performance optimisation for large tree (10,000+ nodes), additional language support, advanced analytics

11. Immediate Next Steps

In order, here is what you need to do first — Claude will guide you through each:

18. Check domain availability for nasab.org / nasab.app and register your preferred domain
19. Create a free Supabase account at supabase.com
20. Create a free Vercel account at vercel.com and connect your GitHub
21. Share your Urdu PDF shajra with Claude to begin digitisation
22. Claude will scaffold the Next.js project and walk you through every file
23. Claude will write and run all database migrations
24. Claude will seed the historical tree data (core 42 generations)
25. We build and deploy the tree MVP together, phase by phase

You do not need to know how to code. Claude will write every line of code, explain every decision, and guide you through every setup step. Your job is to make product decisions, verify the genealogical data, and build the institutional relationships.

وَاللَّهُ وَلِيُّ الْمُؤْمِنِينَ

And Allah is the Guardian of success.